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TEAM NAME: BlockX

COLLEGE NAME: Sri Venkateswara Of Engineering

PROBLEM STATEMENT: Organizations struggle to identify emerging innovation opportunities because valuable insights are hidden within vast and scattered research papers, patents, startup data, and technology news.

PROBLEM STATEMENT

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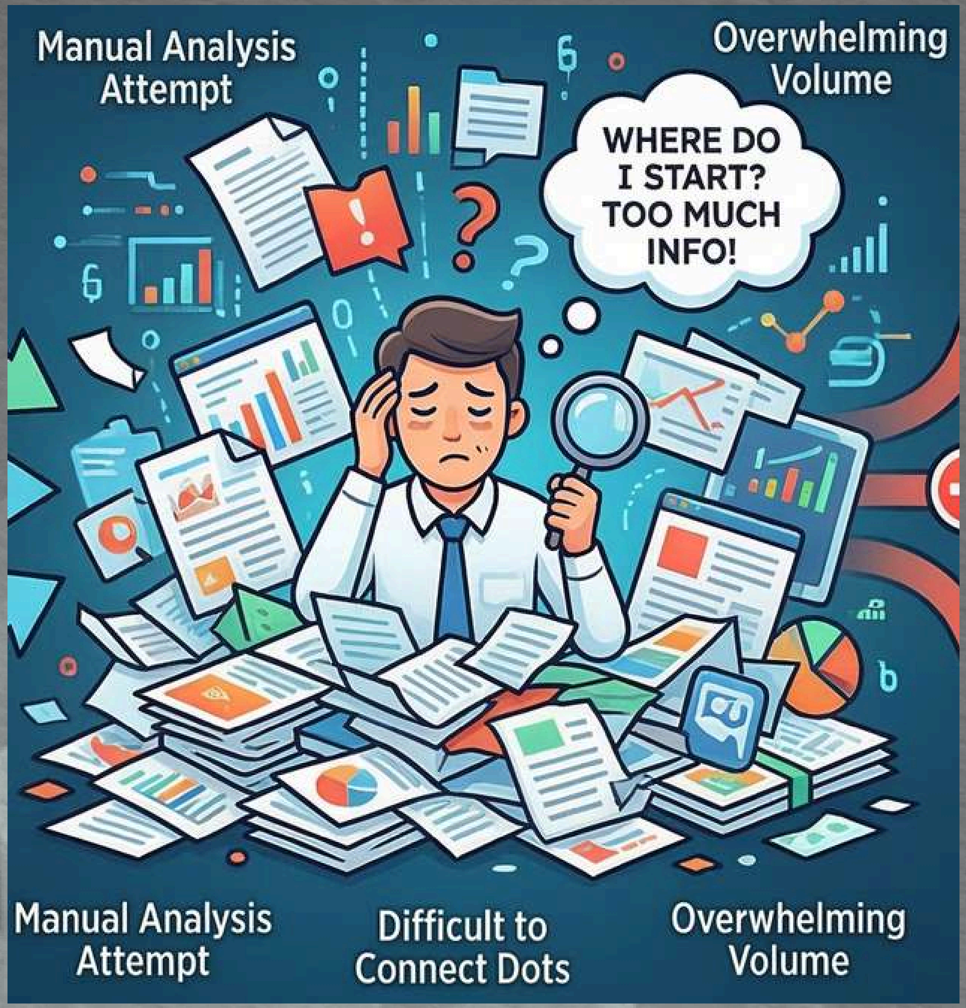

- In today's rapidly evolving technological landscape, companies and research organizations face significant challenges in identifying emerging innovation opportunities. A vast amount of valuable information is continuously generated through research papers, patents, startup activities, and technology news.
- However, this information is scattered across multiple platforms and grows at an overwhelming pace, making it extremely difficult for organizations to manually analyze and extract meaningful insights.
- As a result, many potential innovation opportunities and disruptive technologies remain hidden within these large datasets, causing companies to miss early signals of future technological trends. Therefore, there is a need for an intelligent system that can automatically analyze large-scale technology data, detect emerging trends, and help organizations discover potential innovation opportunities at an early stage.

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SOLUTION

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01 Automated Data Collection

Build a system that gathers data from research papers, patents, startup platforms, and technology news in real time.

02 Technology Evolution Tracking

Monitor how technologies grow over time by analyzing patterns in research publications, patents and startup innovations.

03 AI-Based Trend Detection

Use machine learning and NLP techniques to analyze large volumes of text and identify emerging technology trends.

04 AI-Generated Innovation Insights

Generate potential innovation opportunities and future technology ideas based on detected trends.

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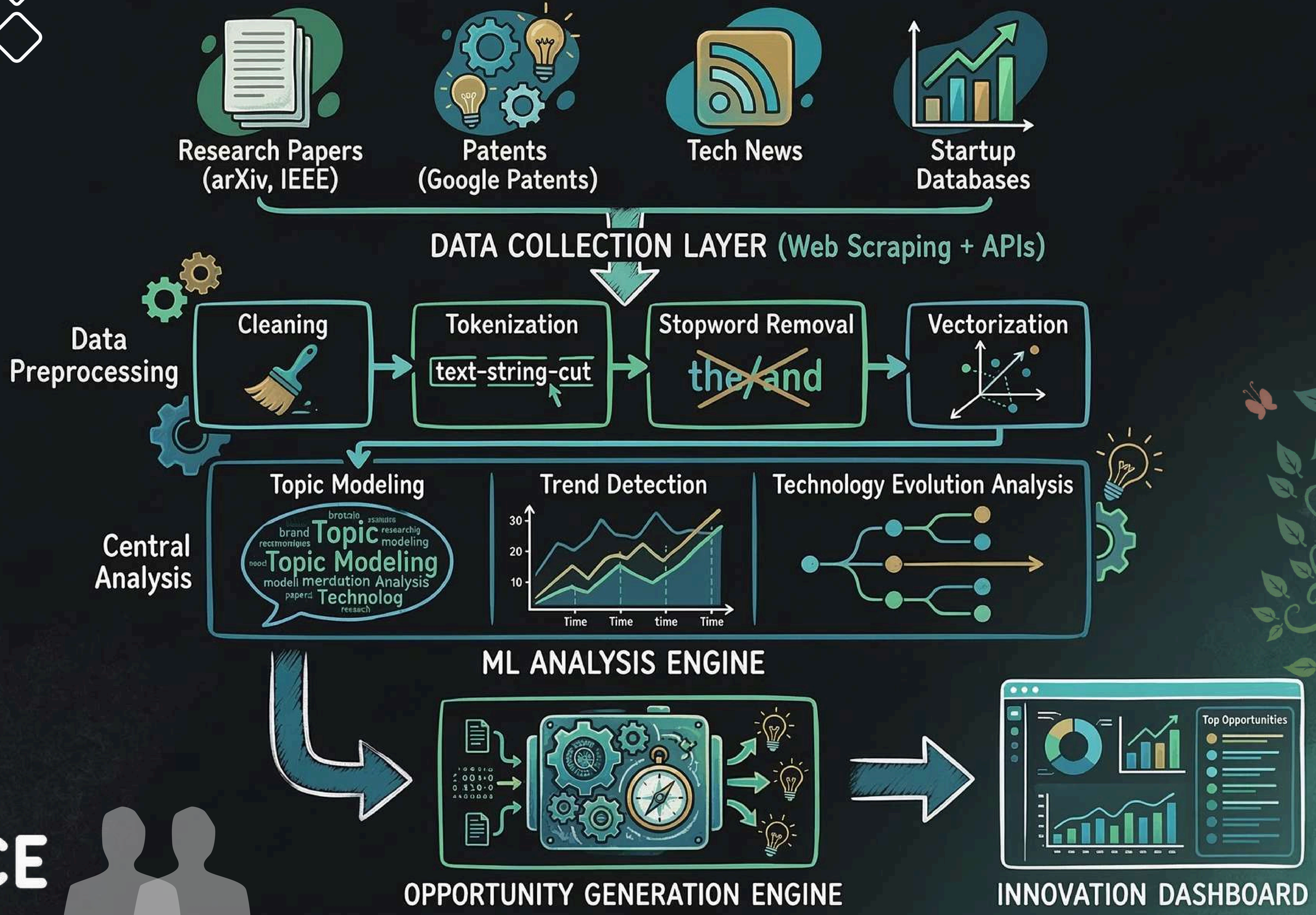
FLOW DIAGRAM



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TECHNICAL REQUIREMENTS

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06 - Machine Learning

Scikit-learn - ML algorithms
TensorFlow / PyTorch - Deep learning models
NLTK - Text preprocessing
spaCy - Natural language processing
Gensim - Topic modeling (LDA)
Sentence Transformers / BERT - Semantic similarity

05 - Backend

- Flask or FastAPI

Purpose:

- Connect ML models with frontend
- Provide APIs for the dashboard

04 - Visualization Tools

Matplotlib - Data visualization
Seaborn - Statistical graphs
Plotly - Interactive charts
Power BI / Tableau - Dashboard visualization

01

02

03

04

05

06

01 - Frontend

React.js - Interactive dashboard
HTML / CSS / JavaScript - Web interface
Streamlit - Quick ML dashboard

02 - Database

MongoDB - Store research papers and text data
PostgreSQL - Structured data storage
Elasticsearch - Fast search of research papers

03 - Data Sources APIs

The system may use:

- arXiv API (Research Papers)
- Google Patents dataset
- TechCrunch API (Tech news)
- Crunchbase API (Startup data)

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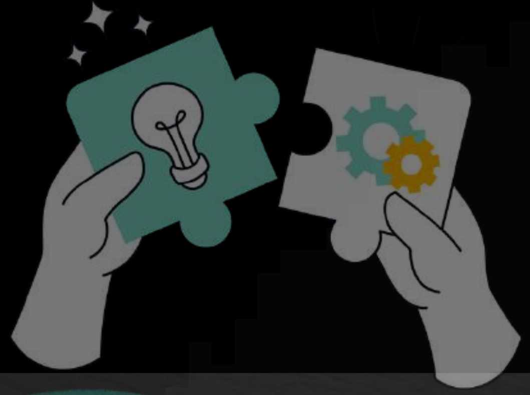
NOVELTY / USE CASE



Early Innovation Detection

Identifies emerging technologies and hidden innovation opportunities before they become mainstream.

Multi-Source Trend Analysis.

Analyzes data from research papers, patents, startups, and tech news to uncover technology trends.

AI-Driven Insights

Generates data-driven innovation ideas to support strategic business and research decisions.

BUSINESS MODEL

Subscription-Based Platform

01

Companies can access the Innovation Discovery Engine through a monthly or yearly subscription model to analyze technology trends and innovation opportunities.

Technology Trend Reports

02

The system generates detailed reports on emerging technologies and innovation trends, which organizations can purchase for research and strategic decision-making.

API and Integration Services

03

Businesses can integrate the innovation analytics APIs into their internal systems to automatically track research trends, patents, and new technology developments.

TARGET MARKET

API and Integration Services

01

Software and AI companies that want to identify emerging technologies and innovation opportunities to stay ahead in the market.

Research Institutions and Universities

02

Research labs and universities that need tools to analyze research trends and identify research gaps for future studies.

Startups and Entrepreneurs

03

Businesses can integrate the innovation analytics APIs into their internal systems to automatically track research trends, patents, and new technology developments.

TEAM DETAILS

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ROLE	NAME	STREAM/DEPT	YEAR	COLLEGE
Team Leader	Nakshatra.L	B.Tech (ADS)	3^{rd} Year	Sri Venkateswara Of Engineering
Team Member	Lathika.S.P	B.Tech (ADS)	3^{rd} Year	Sri Venkateswara Of Engineering
Team Member	Prakash.M	B.Tech (ADS)	3^{rd} Year	Sri Venkateswara Of Engineering
Team Member	Subash Chandra Bose .S	B.Tech (ADS)	3^{rd} Year	Sri Venkateswara Of Engineering
Team Member	Vandana.E	B.Tech (ADS)	3^{rd} Year	Sri Venkateswara Of Engineering
Team Member	Yeseswini.S	B.Tech (ADS)	3^{rd} Year	Sri Venkateswara Of Engineering

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